Toktam Rashidi (she/her)

Department of Physics & Astronomy, McMaster University, Hamilton, ON, Canada

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Research Interests

Computational physics; numerical simulations and modelling of physical systems; astrophysical masers; radio flares and radio transients; quantum coherent phenomena in astrophysics; superradiance; machine learning and optimization.

Education

Ph.D. Astrophysics | McMaster University, Canada — Sep. 2023 – Present
Advisor: Dr. Fereshteh Rajabi; GPA: 3.84/4.0 — *Hamilton, ON, Canada*

M.Sc. Physics | University of Tehran, Iran — 2020 – 2023
Advisor: Dr. Hossein Mohseni Sadjadi; GPA: 3.88/4.0 — *Tehran, Iran*

B.Sc. Physics | University of Tehran, Iran — 2016 – 2020
GPA: 3.82/4.0 — *Tehran, Iran*

Publications and Manuscripts

Peer-Reviewed Publications

1. **Rashidi, T.**, Anari, V., Powles, O., MacLeod, G. C., Tanabe, Y., Yonekura, Y., & Rajabi, F. (2026), *Monthly Notices of the Royal Astronomical Society*, 545(3), staf2200.
   A Unified Maxwell–Bloch Framework for Multiperiodic 6.7 GHz Methanol Flaring in G9.62+0.20E.
2. Fang, N., **Rashidi, T.**, Yonekura, Y., Anari, V., Burns, R., Houde, M., & Rajabi, F. (2026), *New Astronomy*, in press.
   Modelling the 6.7 GHz Methanol Flaring in G358.93−0.03 Using the Maxwell–Bloch Framework.
3. **Rashidi, T.**, Anari, V., Bartkiewicz, A., Wolak, P., Szymczak, M., & Rajabi, F. (2025), *Monthly Notices of the Royal Astronomical Society: Letters*, 542(1), L12–L18.
   Modelling the Periodic 6.7 GHz Methanol Flaring in G22.356+0.066.

Manuscripts under Review

1. Anari, V., **Rashidi, T.**, & Rajabi, F.
   A Race to Coherence: Superradiance and Competing Channels in a Three-Level Λ-System.

Manuscripts in Preparation

1. **Rashidi, T.**, Liu, Z., Anari, V., Yonekura, Y., Bartkiewicz, A., Wolak, P., & Rajabi, F.
   Maxwell–Bloch Modelling of Periodic 6.7 GHz Methanol Flares. I. Stable Quiescent Emission.
2. **Rashidi, T.**, Liu, Z., Anari, V., Yonekura, Y., MacLeod, G. C., & Rajabi, F.
   Maxwell–Bloch Modelling of Periodic Flare Profiles in the 6.7 GHz Methanol Transition. II. Evolving Quiescent Baselines and Complex Flare Structure.

3. Fang, N., **Rashidi, T.**, MacLeod, G. C., Houde, M., & Rajabi, F.
   Modelling the Multitransition Periodic Flaring in G323.46−0.08 Using the Maxwell–Bloch Framework.
4. Liu, Z., **Rashidi, T.**, Anari, V., Yonekura, Y., & Rajabi, F.
   From Maser Amplification to Superradiance: Maxwell–Bloch Modelling of Methanol Flares in G14.230−0.509.

Research Experience

Visiting Researcher | Piwnice Astronomical Observatory — Sep. 2026
Host: Dr. Anna Bartkiewicz — *Toruń, Poland*

- Connected maser monitoring observations to flare modelling, and gave an invited talk on modelling variability in maser lightcurves.

Graduate Researcher | Department of Physics & Astronomy, McMaster University — Sep. 2023 – Present
Advisor: Dr. Fereshteh Rajabi — *Hamilton, ON, Canada*

- Developed one- and three-dimensional superradiance solvers for modelling astronomical radio flares, including transverse effects in coherent emission.
- Developed a numerical package for solving a three-level Maxwell–Bloch system to study parallel transitions.
- Developed Python-based period-analysis pipelines to identify and categorize periodic maser variability.
- Applied superradiance models to maser-hosting star-forming regions, working toward the first superradiance catalogue in astrophysics.

Research Assistant | University of Tehran — Sep. 2020 – Feb. 2023
Advisor: Dr. Hossein Mohseni Sadjadi — *Tehran, Iran*

- Investigated Gauss–Bonnet and mass-varying neutrino couplings and their impact on late-time cosmic expansion and neutrino-mass evolution.

Presentations

Talks

1. **(invited) Modelling Variability in Maser Lightcurves**. Piwnice Astronomical Observatory, Toruń, Poland. 22 September 2026.
2. **A Race to Coherence: Superradiance and Competing Channels in a Λ-System**. 9th Annual Toronto Ultracold Atom Network (TUCAN), University of Toronto, Toronto, ON, Canada. 13 May 2026.
3. **Cooperative Spontaneous Processes and Astronomical Flares**. McMaster Theory Journal Club, McMaster University, Hamilton, ON, Canada. 1 April 2026.
4. **The Effect of Fast Parallel Transitions on a Superradiance Cascade**. McMaster Astronomy Journal Club, McMaster University, Hamilton, ON, Canada. 11 December 2025.
5. **Dicke's Superradiance and Astrophysical Masers: Modelling Flares in G22.356+0.066 and G9.62+0.20E**. Ralposium, McMaster University, Hamilton, ON, Canada. 25–27 June 2025.
6. **Superradiance and Periodic Methanol Flaring in G22.356+0.066**. Physics and Astronomy Graduate Symposium, McMaster University, Hamilton, ON, Canada. 17 October 2024.
7. **Dicke's Superradiance and Periodic Flaring in G22.356+0.066**. Star Formation Workshop, McMaster University, Hamilton, ON, Canada. 12–14 August 2024.

Posters

1. **Unveiling Superradiant Signatures in Periodic Methanol Flares**. CASCA, Montreal, QC, Canada. 9–11 June 2026.
2. **A Unified Maxwell–Bloch Framework to Model Flaring Behaviour in Maser-Hosting Regions**. The 3rd International Online Conference on Universe. 4–6 March 2026.
3. **Dicke's Superradiance in Astrophysical Masers: Modelling Periodic Flaring in G9.62+0.20E**. Women+ in Physics Canada 2025, Queen's University, Kingston, ON, Canada. 22–25 July 2025.
4. **Light-Matter Dynamics Behind Flares in Maser-Hosting Regions**. The 19th Women in Science and Engineering Annual National Research Conference, McMaster University, Hamilton, ON, Canada. 7 March 2025.

Funding & Awards

$5,000 ($\times 2$) · Physics and Astronomy Departmental International Award | McMaster University — 2025 and 2026

$2,950 · School of Graduate Studies Grant in Aid for Research Travel and Field Work | McMaster University — 2025–2026

Direct Admission to M.Sc. Program | University of Tehran; awarded for outstanding undergraduate GPA — 2020

Tuition Waiver for M.Sc. Studies | University of Tehran — 2020–2023

Tuition Waiver for B.Sc. Studies | University of Tehran; awarded for top performance in the Iranian National University Entrance Exam — 2016–2020

Teaching & Mentorship

Research Mentorship

Co-supervisor | Syed Saqib Abbas Rizvi, undergraduate student — Oct. 2026 – Present

Co-supervisor | Zhejian Liu, master's student — Jan. 2026 – Present

Co-supervisor | Oscar Powles, undergraduate student — Sep. 2024 – Jul. 2025

Co-supervisor | Subhradeep Mahata, MITACS student; with Dr. Fereshteh Rajabi and Dr. Vahid Anari — Summer 2024

Teaching Assistantships

Teaching Assistant | McMaster University — Sep. 2023 – Present, *Hamilton, ON, Canada*
Scientific Computing; Introductory Mechanics; Quantum Mechanics II; Waves, Electricity, and Magnetism; Introduction to Modern Physics

Teaching Assistant | University of Tehran — Sep. 2020 – Dec. 2021, *Tehran, Iran*
Advanced Quantum Mechanics; Symmetries and Group Theory; Fundamentals of Elementary Particles; General Physics I

Course Consulting

Course Consultant | Undergraduate Teaching Seminar, McMaster University Winter 2025 & Winter 2026

Advised a student group preparing teaching presentations on particle physics for ISCI 3A12: Light, the Universe, and Everything.

Service, Outreach & Community Engagement

Judge | Canadian Undergraduate Physics Conference (CUPC), McMaster University 16–17 Oct. 2026
Evaluated oral presentations. *Hamilton, ON, Canada*

Chair and Judge | Physics and Astronomy Graduate Symposium Day, McMaster University 15 Oct. 2026
Chaired the astronomy session and evaluated the talks. *Hamilton, ON, Canada*

Event Organizer | Promoting Inclusivity in Physics & Astronomy (PIPA), McMaster University Sep. 2025 – Present
Organize and coordinate student-led events, including coffee hours and movie nights. *Hamilton, ON, Canada*

Presenter | McMaster W. J. McCallion Planetarium Sep. 2025 – Present
Deliver planetarium presentations and support astronomy outreach activities. *Hamilton, ON, Canada*

Astronomy Outreach Facilitator | McMaster University Winter 2025–2026
Hosted kindergarten groups visiting the Planetarium and designed age-appropriate space-themed craft and learning activities. *Hamilton, ON, Canada*

Local Organizing Committee | Toronto Ultra-Cold Atom Network, McMaster University Spring 2024
Supported preparation, logistics, and participant-facing organization for a professional scientific meeting. *Hamilton, ON, Canada*

Course Assistant and Organizer | General Physics, Remote Format, University of Tehran Fall 2020
Supported the transition of an introductory physics course to an online format and designed online quizzes and assessment tools. *Tehran, Iran*

Organizer and Facilitator | Weekly Feminist Reading Circle, University of Tehran Winter 2019 – Spring 2020
Initiated and coordinated weekly reading sessions and selected reading materials. *Tehran, Iran*

Event Organizer | Alumni–Student Gathering, University of Tehran Winter 2017, Winter 2018, Winter 2019
Helped plan and coordinate an annual departmental event bringing together alumni, current students, and faculty. *Tehran, Iran*

Skills

Scientific computing: Python; C; C++; MAPLE
Research methods: Numerical analysis; data analysis; mathematical modelling
Languages: Persian: native; English: fluent; French: elementary